Updated November 11, 2026

Homework problems for AMAT 220 (Linear Algebra), Fall 2026. Over the course of the semester I'll add each week's homework set to this pdf, with each due date specified. Each problem is worth 2 points.

Homework set #1 (five problems) (due Weds 9/2).

Problem 1.1: Write down the augmented matrix encoding the system of linear equations

$$\begin{aligned} 2x - 3y + 5z &= 7 \\ x - z &= 1 \end{aligned}$$

Problem 1.2: Starting with the matrix $\begin{pmatrix} 0 & 3 & -4 & 8 \\ -1 & -1 & 2 & -4 \end{pmatrix}$, write down the matrix obtained from the elementary row operation that replaces the first row with itself plus 2 times the second row.

Problem 1.3: Explain in words why the system of linear equations

$$\begin{aligned} x + y &= 2 \\ x - y &= 0 \\ x + 3y &= 5 \end{aligned}$$

has no solutions. (It's fine to not use matrices.)

Problem 1.4: Exactly one of the following three statements is true, and the other two are false. Figure out which, explain why it's true, and explain why the other two are false.

(1) The equations $x + y + z = 0$ and $2x + 2y + 2z = 0$ have no solutions in common.
(2) The equations $x + y + z = 0$ and $2x + 2y + 2z = 0$ have exactly one solution in common.
(3) The equations $x + y + z = 0$ and $2x + 2y + 2z = 0$ have infinitely many solutions in common.

Problem 1.5: Is the matrix $\begin{pmatrix} 1 & 0 & 0 & 8 & 3 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$ in reduced row echelon form or not? Justify your answer.

No class Labor Day 9/7

Homework set #2 (five problems) (due Weds 9/9).

Problem 2.1: Use the official row reduction algorithm to put the matrix $\begin{pmatrix} 0 & 1 & -1 & 2 \\ 1 & 0 & -3 & 0 \\ 2 & -2 & 1 & 1 \end{pmatrix}$ in reduced row echelon form.

Problem 2.2: Use the official algorithm to find all solutions to the system of linear equations

$$\begin{aligned} y - z + 2w &= 2 \\ x - 3z &= 1 \\ 2x - 2y + z + w &= 3 \end{aligned}$$

Problem 2.3: Determine whether or not the system of linear equations

$$\begin{aligned} y - z &= 2 \\ x - 3z &= 1 \\ 2x - 2y - 4z &= 0 \end{aligned}$$

is consistent. Justify your answer.

Problem 2.4: Rewrite the vector equation $x \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} + y \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + z \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$ as a system of linear equations.

Problem 2.5: Find all solutions to the vector equation in Problem 2.4.

Homework set #3 (five problems) (due Weds 9/16).

Problem 3.1: Is the vector $\begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$ in the span of $\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \right\}$? Justify your answer.

Problem 3.2: Do the vectors $\left\{ \begin{pmatrix} 4 \\ -8 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \end{pmatrix} \right\}$ span all of $\mathbb{R}^2$? Justify your answer.

Problem 3.3: Write down the vector equation whose corresponding system of linear equations is

$$\begin{aligned} 3x - y + 2z &= 4 \\ -x + 5z &= -2 \\ 2x + 6y &= 0 \end{aligned}$$

Problem 3.4: Write down the matrix equation corresponding to the system of linear equations from Problem 3.3. (The matrix equation, not the augmented matrix.)

Problem 3.5: Consider the homogeneous matrix equation $A\vec{x} = \vec{0}$ where $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$. Find a non-trivial solution. (Just one non-trivial solution is fine, you don't need to describe all of them.)

Reminder: Exam in class on Weds 9/23.

Homework set #4 (five problems) (due Weds 9/23).

Problem 4.1: Describe the set of solutions to $6x - 6y + 3z = 0$ as the span of two vectors.

Problem 4.2: Consider the non-homogeneous matrix equation $A\vec{x} = \vec{b}$ where $A = \begin{pmatrix} 5 & -6 & 11 \\ 4 & 7 & -2 \\ 3 & 8 & -1 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 10 \\ 9 \\ 10 \end{pmatrix}$. Find a non-trivial solution. (Just one non-trivial solution is fine, you don't need to describe all of them; and for once I actually don't recommend trying to row reduce.)

Problem 4.3: Consider the non-homogeneous matrix equation $A\vec{x} = \vec{b}$ where $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$. Describe the set of solutions in parametric vector form.

Problem 4.4: Determine whether the vectors $\left\{ \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} \right\}$ are linearly independent or linearly dependent.

Problem 4.5: Show that the vectors $\left\{ \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \right\}$ are linearly dependent, and find an explicit non-trivial linear combination of them that equals $\vec{0}$.

Homework set #5 (five problems) (due Weds 9/30).

Problem 5.1: Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation that reflects $\vec{x}$ across the "$y = -x$ line" in the plane (the line going from the top-left, through the origin, to the bottom-right). Compute the image of $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ under T.

Problem 5.2: Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation that rotates the plane 90 degrees counterclockwise around the origin. Find $\vec{x}$ in $\mathbb{R}^2$ such that T sends $\vec{x}$ to $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$.

Problem 5.3: Let $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation that sends $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} y \\ 3x - 2y \\ x + y \end{pmatrix}$ (trust me this is a linear transformation). Is $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ in the image of T?

Problem 5.4: Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation $T(\vec{x}) = A\vec{x}$, where $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Solve $T(\vec{x}) = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$ for $\vec{x}$.

Problem 5.5: Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation $T(\vec{x}) = A\vec{x}$, where $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}$. Find all solutions to $T(\vec{x}) = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$, in parametric vector form.

Homework set #6 (five problems) (due Weds 10/7).

Problem 6.1: Let $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation that sends $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} y \\ 3x - 2y \\ x + y \end{pmatrix}$. Compute the standard matrix for T.

Problem 6.2: Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ be the projection to the xy-plane, so T sends $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to $\begin{pmatrix} x \\ y \end{pmatrix}$. Determine whether or not T is one-to-one, and whether or not T is onto. [You can use matrices if you want, but it might be easier to just work with the definitions of one-to-one and onto.]

Problem 6.3: Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation that sends $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to $\begin{pmatrix} x + y + z \\ -2x + y \\ 3x + y - z \end{pmatrix}$. Determine whether or not T is one-to-one, and whether or not T is onto. [Now you definitely want to use matrices.]

Problem 6.4: Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. Compute the product AAA.

Problem 6.5: For a variable x, let $A = \begin{pmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and let $B = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Determine all the values of x that make A and B commute.

Homework set #7 (five problems) (due Weds 10/14).

Problem 7.1: Let $A = \begin{pmatrix} 1 & x \\ 3 & 6 \end{pmatrix}$ for x a variable. Determine all the values of x for which A is invertible.

Problem 7.2: Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ -2 & -3 & -4 \end{pmatrix}$. Determine whether A is invertible or not. (Try to do as little work as possible!)

Problem 7.3: Let $A = \begin{pmatrix} 1 & -3 & 7 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{pmatrix}$. Compute the inverse of A.

Problem 7.4: Give an example of two 2-by-2 matrices A and B such that A and B are invertible but $A + B$ is not invertible (and justify your answer).

Problem 7.5: Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Suppose that $T(\vec{e}_1)+\cdots+T(\vec{e}_n) = \vec{0}$. Determine whether or not T is is onto. [Oops I meant $m = n$, as written this one was "not enough information to tell".]

No class on Weds 10/21 (I'm out of town), and no homework due that day. The next homework will be due Weds 10/28, on the day of the exam.

Reminder: Exam in class on Weds 10/28.

Homework set #8 (five problems) (due Weds 10/28).

Problem 8.1: Compute the cofactor expansion of $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ -2 & -3 & -4 \end{pmatrix}$ across the second row (and then use it to compute the determinant).

Problem 8.2: Compute the determinant of $A = \begin{pmatrix} 1 & -31 & 0 & 26 \\ 0 & 13 & 0 & 500 \\ -41 & 85 & 2 & 99 \\ 0 & -2 & 0 & 1 \end{pmatrix}$ (try to do as little work as possible).

Problem 8.3: For x a variable, let $A = \begin{pmatrix} 1-x & 2 \\ 2 & -2-x \end{pmatrix}$. Determine which values of x make A non-invertible.

Problem 8.4: Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, d > 0$ and $c < 0$. Explain why A must be invertible, even without knowing anything else.

Problem 8.5: Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. Suppose that $T(\vec{e}_1)+\cdots+T(\vec{e}_n) = \vec{0}$. Compute $\det(T)$ (and justify your answer).

Homework set #9 (five problems) (due Weds 11/4).

Problem 9.1: Let S be the set of all $\begin{pmatrix} x \\ y \end{pmatrix}$ such that $y = x^2$. Show that S is not a subspace of $\mathbb{R}^2$.

Problem 9.2: Let V be the vector space of all functions $f\colon \mathbb{R} \to \mathbb{R}$. Let $T\colon V \to \mathbb{R}$ be the function $T(f(x)) = f(7)$. Show that T is a linear transformation.

Problem 9.3: Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x-y+2z \\ x+y+4z \\ 2y+4z \end{pmatrix}$. Describe the kernel of T as a span of vector(s).

Problem 9.4: Using the same T as in Problem 9.3, determine whether $\begin{pmatrix} 3 \\ 7 \\ 6 \end{pmatrix}$ lies in the image of T.

Problem 9.5: Let V be a vector space and $T\colon V \to W$ an invertible linear transformation with inverse $T^{-1}\colon W \to V$ (meaning $(T^{-1} \circ T)(\vec{v}) = \vec{v}$ and $(T \circ T^{-1})(\vec{w}) = \vec{w}$ for all $\vec{v}$ in V and $\vec{w}$ in W). Show that T^{-1} is also a linear transformation; in other words, show that $T^{-1}(a\vec{w} + \vec{w}') = aT^{-1}(\vec{w}) + T^{-1}(\vec{w}')$ for all $\vec{w}$ and $\vec{w}'$ in W and all a in $\mathbb{R}$.

Homework set #10 (five problems) (due Weds 11/11).

Problem 10.1: Let $A = \begin{pmatrix} 1 & 3 & 2 & 1 \\ -2 & -6 & -3 & 0 \end{pmatrix}$. Find bases for the null space Null(A) and the column space Col(A) of A.

Problem 10.2: Let V be a vector space and $T\colon V \to V$ a linear transformation from V to itself. Assume that image(T) = ker(T). Explain why the only way T can be one-to-one and onto is if V is nothing but $V = \{\vec{0}\}$.

Problem 10.3: Let P_2 be the vector space of polynomials with degree at most 2. Let $S = \{3 + 5x + 7x^2, -2 + 4x - 9x^2\}$. Is S a basis for P_2? Justify your answer.

Problem 10.4: Let P_2 be the vector space of polynomials with degree at most 2. Let $\mathcal{B} = \{1 + x + x^2, 2 - 2x^2, 2x + 5x^2\}$ (trust me this is a basis). Compute $[4 + 6x + 2x^2]_{\mathcal{B}}$.

Problem 10.5: Let V be the span of $v_1 = \begin{pmatrix} 1 \\ 1 \\ 6 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 2 \\ 3 \\ 9 \end{pmatrix}$. Let $\vec{b} = \begin{pmatrix} 3 \\ 5 \\ 12 \end{pmatrix}$ (trust me this is in V). Compute $[\vec{B}]_{\{\vec{v}_1, \vec{v}_2\}}$.

Homework set #11 (five problems) (due Weds 11/18).

Problem 11.1: Compute the dimension of the subspace of $\mathbb{R}^4$ spanned by $\left\{ \begin{pmatrix} 1 \\ 2 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 6 \\ -4 \\ 11 \end{pmatrix} \right\}$.

Problem 11.2: Let $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 5 & 8 \\ -2 & 0 & 2 \\ 1 & 2 & 3 \end{pmatrix}$. Compute the rank and nullity of A, and verify that the rank-nullity theorem holds.

Problem 11.3: Let $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 4 \\ 7 \end{pmatrix} \right\}$ and $\mathcal{C} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$ be bases for $\mathbb{R}^3$. Compute the change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$, and use it to compute the coordinate vector $\left[\begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix} \right]_{\mathcal{C}}$.

Problem 11.4: In the vector space $V = P_1$, let $\mathcal{B} = \{4-3x, -5+4x\}$ and $\mathcal{C} = \{5+7x, 7+10x\}$ be bases. Compute the change of coordinates matrices from $\mathcal{B}$ to $\mathcal{C}$, and from $\mathcal{C}$ to $\mathcal{B}$, and use the appropriate one to compute $[7 + 10x]_{\mathcal{B}}$.

Problem 11.5: Compute the eigenvalues of $\begin{pmatrix} 7 & 3 \\ -1 & 3 \end{pmatrix}$. (Show your work.)

No class on Weds 11/25 (Thanksgiving Eve).

Reminder: Exam in class on Weds 12/2.

Final homework will be due Weds 12/2.
